

GROWRIDER · DESIGN DECISION RECORD

GrowRider recommendations accepted

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DOCUMENT CONTROL

Document ID	GRR-DDR-002	Version	0.2
Status	Draft	Date	2026-09-26
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/growrider		
Source	docs/decisions/0002-recommendations-accepted.md @ d0c0d47		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-26	Amish Chadha	Budget top-up to \$300 decided by Amish (N1)
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted for items D7 to D13; D1 to D6 confirmed; items O1 to O4 and N1 to N2 remain proposed

■ Context

The TRL 3 review note (docs/REVIEW.md , session 2026-09-25: TRL 3) listed seven new items as "Proposed, awaiting Amish", each with a recommendation, and kept four TRL 2 items open because no recommendation had been made. GRR-DDR-001 had recorded D1 to D6 as "Adopted as recommended" under Amish's instruction of the same day. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item with a recommendation is therefore decided in favor of that recommendation; where several options were offered, the recommended option is the decision. Items without a recommendation stay open. TRL 4 remains on hold by Amish's instruction, so any decided item that needs building, testing or purchasing is recorded as decided but on hold.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25: TRL 3) and in GRR-CAL-001 v0.1.

■ Decision

Table 1. Items decided by this record.

#	Item	Decision	What changed in the repo
D1 to D6	Wheel size, brakes, frame, tires, crank length, \$120 production target (GRR-DDR-001)	Decided by Amish, 2026-09-25: go with recommendation (confirmed)	Wording in GRR-DDR-001 and REVIEW unchanged in substance
D7	Mass (R5)	Decided by Amish, 2026-09-25: go with recommendation. Option (b): chromoly main tubes, an aluminium rack, alloy rims and an alloy bar, keeping solid tires (D4); R5 stays at 13 kg as the production goal	cad/src/model.py : seat tube 31.8 x 1.2 mm, down tube 31.8 x 0.9 mm and loop tube 28.6 x 0.9 mm in chromoly 4130; rack 12 x 1.5 mm aluminium 6061-T6. BOM lines 1, 6, 8 and 13 respecified and repriced. GRR-CAL-001 v0.2: mass 15.3 kg to 14.4 kg (the TRL 3 note estimated 13.6 kg; the model-based figure is higher because the seat tube keeps a 1.2 mm wall for the sleeve and the head tube keeps 1.5 mm). R5 restated as the production goal; still not met
D8	Alloy front rim (R7)	Decided by Amish, 2026-09-25: go with recommendation. Alloy front rim specified	BOM line 7 now alloy only; R7 target names the alloy rim; wet front braking for the smallest rider about 0.15 g instead of 0.05 g with steel
D9	Coaster hub brake data (R7)	Decided by Amish, 2026-09-25: go with recommendation. Obtain the brake ratio of a regional coaster hub before further brake work	Recorded as an open action; the data must come from a hub maker or the co-design partner. Any hub purchase or bench test is TRL 4 work, on hold
D10	Strength (R12)	Decided by Amish, 2026-09-25: go with recommendation. Chromoly steerer and quill, and a 1.8 mm wall sleeve; recheck in GRR-CAL-001	Sleeve 28.6 x 1.5 mm became 29.2 x 1.8 mm chromoly (25.6 mm bore for the 25.4 mm post), which needs the 29.4 mm bore of a 1.2 mm seat tube. Steerer and quill chromoly at the same size. GRR-CAL-001 v0.2 adds a 90 MPa screen for chromoly sections: 3 of 6 sections above the screen became 1 of 6 (steerer at the crown, 117 MPa). Drawing GRR-DWG-001 to Rev P2
D11	R2 target	Decided by Amish, 2026-09-25: go with recommendation. R2 widened from 436 to 654 mm to 400 to 670 mm	GRR-REQ-001 v0.4; the design already meets it
D12	R9 tools	Decided by Amish, 2026-09-25: go with recommendation. R9 split into a parent tool list (13 mm spanner, screwdriver) and a mechanic tool list (13, 15 and 32 mm spanners, BB lockring spanner, screwdriver)	GRR-REQ-001 v0.4 (R9a, R9b); both tool lists met on paper; the 10 min fit change sits at the limit, so R9 stays at risk
D13	Low bar option	Decided by Amish, 2026-09-25: go with recommendation. A flat bar for the smallest riders is a co-design question	Added to the open questions of GRR-PRB-001 v0.4 and GRR-PRC-001 v0.4; no design change

Table 2. Items still open.

#	Item	Status
O1	Helmet funding (BOM line 20, \$12)	Proposed, awaiting Amish (no recommendation was made). Since the 2026-09-26 top-up (N1), the \$293 total with the helmet fits the \$300 budget
O2	First partner	Proposed, awaiting Amish; community designs pick co-design partners per area later
O3	First region (Zambia, Kenya or Malawi suggested)	Proposed, awaiting Amish (no recommendation was made)
O4	Co-design partner for fit measurements and the regional parts list	Open; picked per area later
N1	Prototype budget: D7 and D10 raise the bike to \$281 (\$293 with the helmet) against the \$250 in <code>project.yaml</code>	New, proposed, awaiting Amish. Options: raise the budget to \$300; keep \$250 and build the first prototype in hi-tensile steel with the chromoly parts later; or keep \$250 and treat R11 as not met. Recommendation: raise to \$300, which also settles O1. Budget top-up to \$300: decided by Amish, 2026-09-26. <code>budget_usd</code> is now \$300; GRR-REQ-001 v0.5 and GRR-CAL-001 v0.3 record R11 as at risk (prototype \$281, \$293 with the helmet, within budget; production cost not estimated)
N2	Steerer at the crown still above its screen (117 MPa against 90 MPa; 98 MPa with a 25.4 x 2.0 mm steerer)	New, proposed, awaiting Amish. Recommendation: specify a 25.4 x 2.0 mm butted chromoly steerer and confirm against the ISO 4210-2 fork tests at TRL 4

■ Consequences

- GRR-REQ-001, GRR-PRC-001 and GRR-PRB-001 go to v0.4 and GRR-CAL-001 to v0.2; drawing GRR-DWG-001 goes to Rev P2; the STEP, STL and media files are regenerated from the model.
- Requirement status after this record (GRR-CAL-001 v0.2): 5 met, 2 not met (R5 mass 14.4 kg; R11 prototype cost \$281), 3 at risk (R7, R9, R12), 2 not verifiable at TRL 3. See GRR-CAL-001, Table 6.
- `trl` and `trl_target` stay at 3. TRL 4 is on hold by Amish's instruction; nothing in this record authorizes building, testing or purchasing.